INCH-POUND

MIL-R-23417/4A
25 April 1991

SUPERSEDING
MIL-R-23417/4
19 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 15R2N4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Excitation winding	Rotor		
Rated frequency	400	Hz	±1%
Input voltage:			
Maximum	26	volts (rms)	±1%
Test	20	volts (rms)	±1%
Minimum	0.5	volts (rms)	±1%
Primary current	111	milliamps	nominal
Primary power	0.46	watts	nominal
D.C. resistance:		ohms	
Rotor	22.7		±15%
Stator	4.6		±15%
Impedance:			
Zro	37+j175	ohms	R±15%+jx±10%
Zso	8+J40	ohms	R±15%+jx±10%
Zss	11+j4	ohms	Nominal
Transformation ratio:			
Output/input	0.453		±1%
Equality of transformation ratios:			
Output/input	0.08	percent spread	maximum

AMSC - N/A

FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Phase shift:			
Output/input	7.0	degrees	±1.0
Electrical error	3.0	minutes	maximum
Null voltage:			
Fundamental	5.0	mV(rms) output	maximum
Total	9.0	mV(rms) output	maximum
Shift of resolver zero with frequency	0.05	minutes	maximum
Harmonic distortion	0.1	percent	maximum
Temperature rise	15	degrees C	maximum
Mechanical breakaway torque:	0.15	oz-in	maximum
@ -55°C	0.30	oz-in	maximum
Radial play	0.0005	inch	maximum
End play	0.0010	inch	±0.0005
Resonant frequency	250	kHz	minimum
Low corner frequency	46	Hz	maximum

NOTE:

1. A voltage value of 0.5 to 26 volts shall be entered in the "volts" designation space on the identification plate.

TABLE II. Part identifying number variant characteristics

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/4-01A	.555	.500	2.100	Spline	Terminal Board	.1872
M23417/4-02A	.305	.250	1.850	Plain	Wire Leads	.1200
M23417/4-03A	.492	.437	1.850	Plain	Wire Leads	.1200
M23417/4-04A	.555	.500	1.850	Plain	Wire Leads	.1200
M23417/4-05A	.305	.250	2.100	Plain	Terminal Board	.1200
M23417/4-06A	.492	.437	2.100	Plain	Terminal Board	.1200
M23417/4-07A	.555	.500	2.100	Plain	Terminal Board	.1200

1/ Part identifying number M23417/4-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

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TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-R-23417/4	
M23417/4-01	M23417/4-01A
M23417/4-02	M23417/4-02A
M23417/4-03	M23417/4-03A
M23417/4-04	M23417/4-04A
M23417/4-05	M23417/4-05A
M23417/4-06	M23417/4-06A
M23417/4-07	M23417/4-07A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Preparing activity:
Army - AR

(Project 5990-0377)

Review activities:
Navy - OS, EC
Air Force - 17, 99
DLA - ES

User activities:
Army - MI, AV, ME, AT
Navy - CG, MC
Air Force - 14, 19

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